Tata Power Smart Energy Control 360°

POWER BI & MYSQL ANALYTICS PROJECT

Plants · Grid · Smart Meters · EV Mobility · Risk Intelligence

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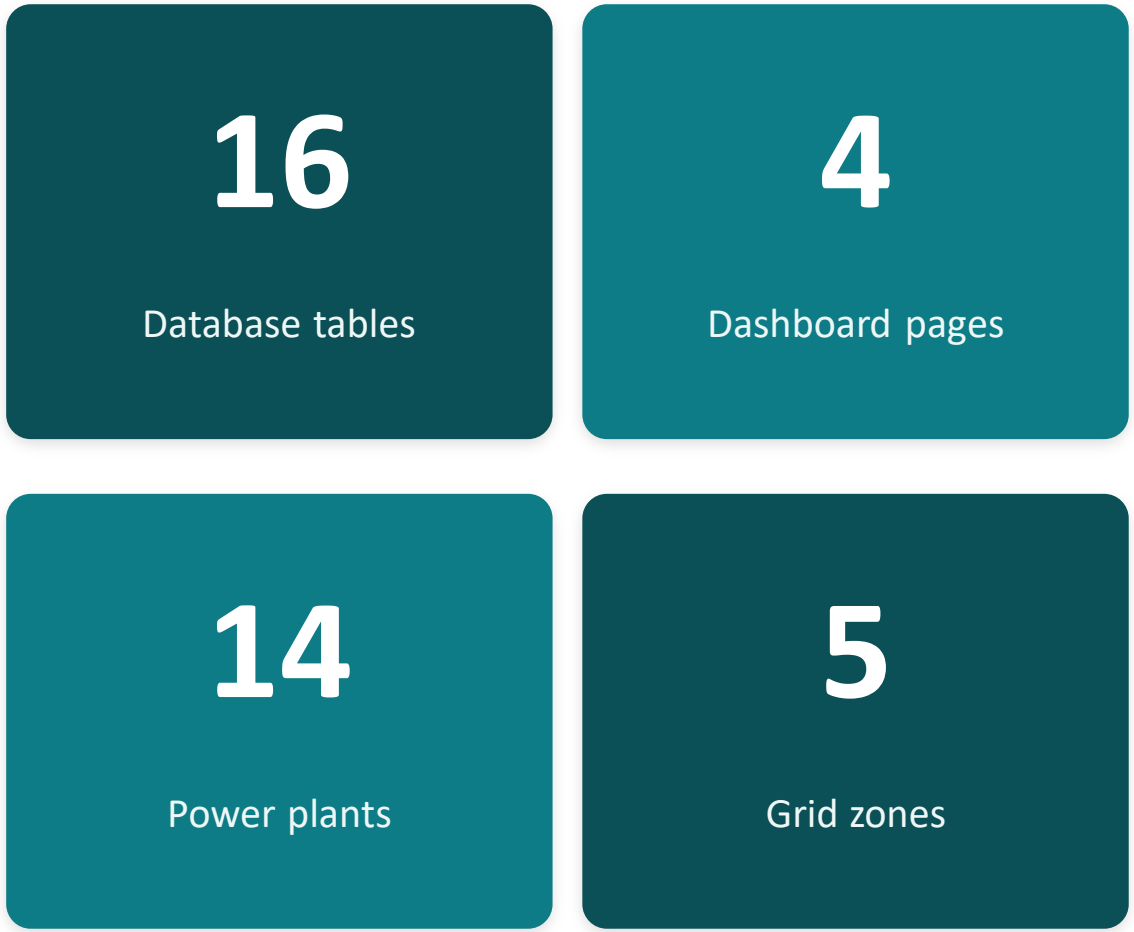
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Project Overview

Tata Power Smart Energy Control 360° is a real-time style analytics solution that connects MySQL to Power BI.

- Brings plants, grid zones, substations, smart meters, EV charging and billing into one view
- Covers 6 fuel types: thermal, hydro, solar, wind, gas and pumped storage
- Covers 5 zones: Ajmer, Delhi, Jamshedpur, Mumbai and Odisha
- Tracks generation mix, load factor, outages and alerts
- Helps teams spot risk early and act faster



Problem & Objectives

The Challenge

- Plant, grid, billing and EV data sit in separate places
- Outages and grid stress are hard to spot early
- No single view of generation, demand, supply and revenue
- Readings arrive every 15 minutes, too many to review by hand
- Manual reporting slows decisions

Our Objectives

- Build one MySQL data model of 16 tables
- Monitor plants and grid health live
- Analyse billing, smart meters and EV charging revenue
- Flag outage, substation and alert risk in one place
- Track CO2 and clean energy share across the fleet

Challenges & Solutions

1	**Blank sensor values in CSV files**	NULLIF turns blanks into NULL during the load
2	**Duplicate time-stamped readings**	Composite primary keys (Timestamp + ID)
3	**Slow time-based queries**	Indexes on plant, zone and alert time
4	**Dashboards need the latest status**	Views such as v_plant_latest and v_open_alerts
5	**Many related tables to manage**	Star schema: 6 dimensions and 10 facts

The Analytics Stack

SQL

MySQL

Structured relational storage across 16 linked tables

ETL

Power Query

Data preparation and transformation layer

PBI

Power BI

Modelling, DAX calculations, KPIs & interactive visuals

DAX

DAX

Calculated measures & columns behind every KPI

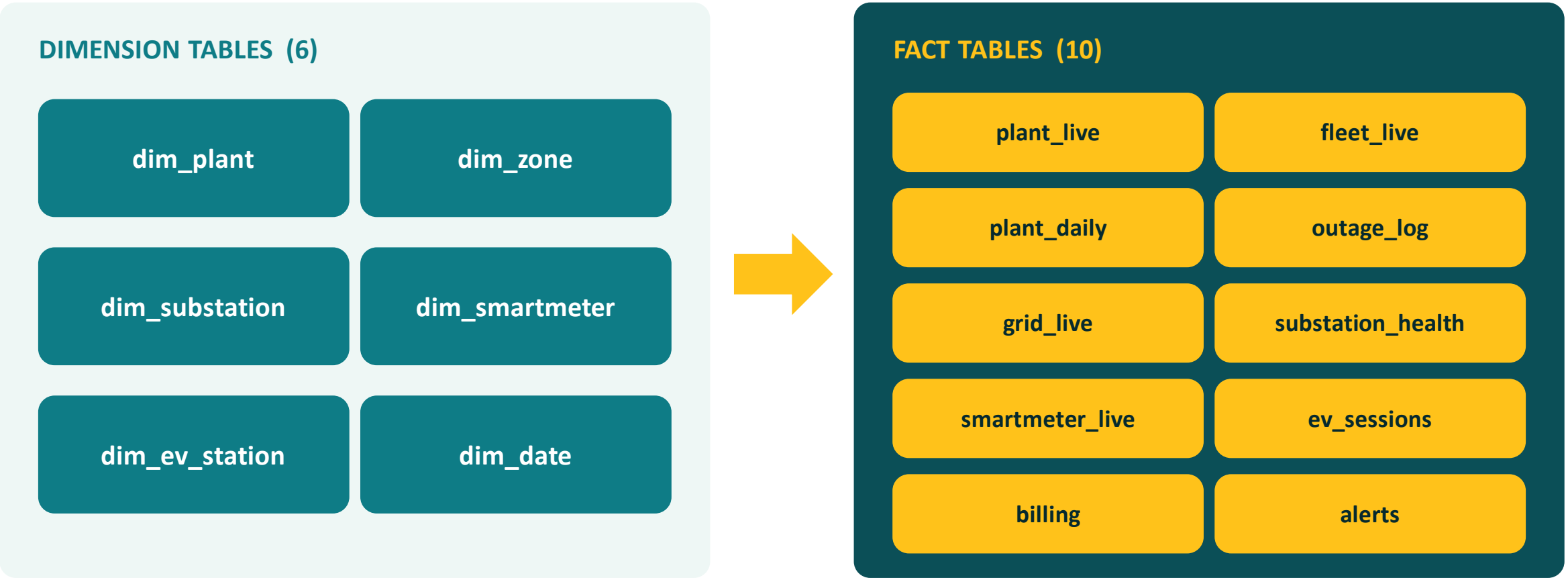
GIT

GitHub

Version control for PBIX, scripts & documentation

Ready for live refresh: indexes and views such as v_plant_latest, v_fleet_latest and v_open_alerts are built for DirectQuery and time filters.

Data Model: 6 Dimensions, 10 Facts



Every table has a primary key. Live fact tables use a composite key of Timestamp + ID.

SQL: Database & Dimension Table

1. Database and dimension table: dim_plant

```sql
-- Tata Power Smart Energy Control 360 - MySQL schema (MySQL 8.x)
CREATE DATABASE IF NOT EXISTS tata_power_360 CHARACTER SET utf8mb4;
USE tata_power_360;

DROP TABLE IF EXISTS dim_plant;
CREATE TABLE dim_plant (
  `PlantID` VARCHAR(50) NOT NULL,
  `Plant_Name` VARCHAR(100) NOT NULL,
  `Fuel_Type` VARCHAR(50) NOT NULL,
  `Segment` VARCHAR(50) NOT NULL,
  `Region` VARCHAR(50) NOT NULL,
  `State` VARCHAR(50) NOT NULL,
  `Latitude` DOUBLE NOT NULL,
  `Longitude` DOUBLE NOT NULL,
  `Capacity_MW` INT NOT NULL,
  `Ownership` VARCHAR(50) NOT NULL,
  `Base_PLF` DOUBLE NOT NULL,
  `Emission_Factor_tCO2_per_MWh` DOUBLE NOT NULL,
  `Temp_Limit_C` INT NOT NULL,
  `Data_Note` VARCHAR(100) NOT NULL,
  PRIMARY KEY (`PlantID`)
);
```

What it does

- Creates the project database in UTF-8
- DROP TABLE IF EXISTS lets the script re-run cleanly
- dim_plant describes each of the 14 plants
- Holds fuel type, region, state, capacity and location
- Base_PLF and emission factor are stored per plant
- PlantID is the primary key used in joins

Cleaning, Visualization & DAX

Data Cleaning

- NOT NULL rules and primary keys block bad rows
- Blank sensor values converted to NULL with NULLIF
- Correct types: DATETIME, DATE, DOUBLE, INT
- Tables named Dim_ and Fact_ in Power BI
- Relationships checked on ZoneID, PlantID, StationID

Data Visualization

- KPI cards for headline numbers
- Donut, treemap, waterfall and combo charts
- Maps for plants, zones and EV stations
- Slicers: state, zone, date range, severity
- Dark Tata-themed design with hover details

DAX Measures

- Total Billed = SUM(Fact_Billing[Amount_INR])
- Collection Efficiency % = DIVIDE(Paid, Total Billed)
- Active Smart Meters = DISTINCTCOUNT of meters
- EV Sessions = COUNT of SessionID
- EV Energy kWh = SUM(kWh_Delivered)

Dashboard 1 · Plant Overview



Page 1 of 4: Smart Energy Control 360°

Takeaway: Availability is 100% but PLF is 27.77%, so there is room to raise utilisation.

KEY KPIs

Total Output: **73.73K MW**
Plant Load Factor: **27.77**
Avg Efficiency: **31.11**
Avg Availability: **100.00**
Total Revenue: **168.41**

VISUALS

- Generation mix across 14 plants
- Plant location and output
- Output trend treemap and Top 5 plants
- Live plant status with state slicer

Dashboard 2 · Grid & Substations



Page 2 of 4: Grid and Substations

Takeaway: Frequency of 49.99 Hz and a grid health score of 89.20 point to a stable grid.

KEY KPIs

Grid Frequency: **49.99 Hz**
Peak Demand: **568.60 MW**
Unserved Energy: **167.83K**
Technical Loss: **6.08%**
Substation Load: **49.40**

VISUALS

- Demand vs supply and unserved energy by zone
- Grid health score: 89.20
- Top 10 substations and grid status
- Slicers for faults, zone and temperature

Dashboard 3 · Billing & EV Mobility



Page 3 of 4: Billing & EV Mobility

Takeaway: Residential bills dominate, and 15K EV sessions show strong charging demand.

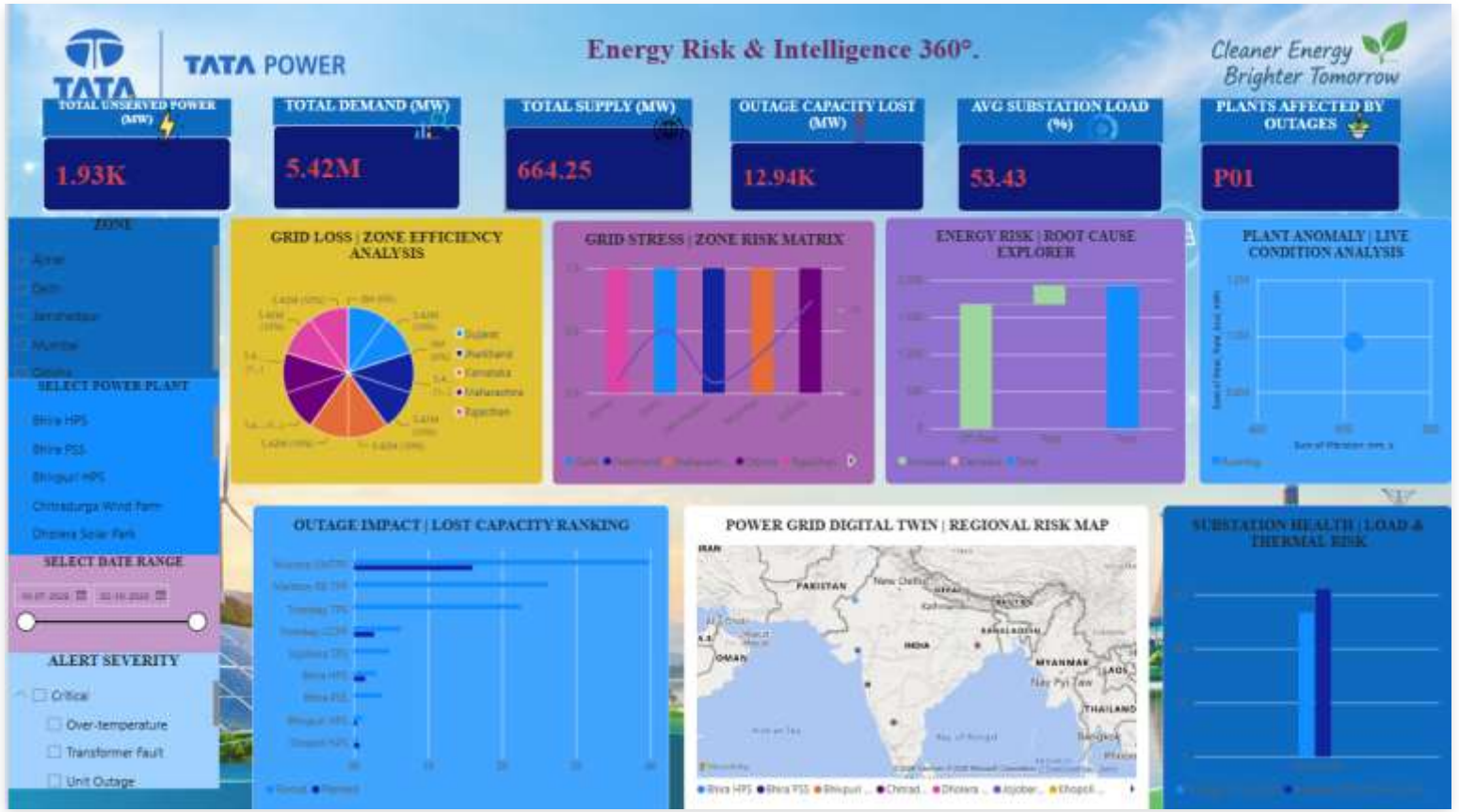
KEY KPIs

Total Billed: **37.45K INR**
Units: **1.72K kWh**
Active Smart Meters: **5K**
EV Charging Sessions: **15K**
Rooftop Solar: **5K**

VISUALS

- EV revenue by zone: AC vs DC fast
- Billing payment status by category
- Monthly billing revenue trend
- Meter reading status and billing alerts

Dashboard 4 · Energy Risk 360°



Page 4 of 4: Energy Risk & Intelligence 360°

Takeaway: Mundra UMTPP leads lost capacity; Trombay TPS (P01) is the plant flagged as affected.

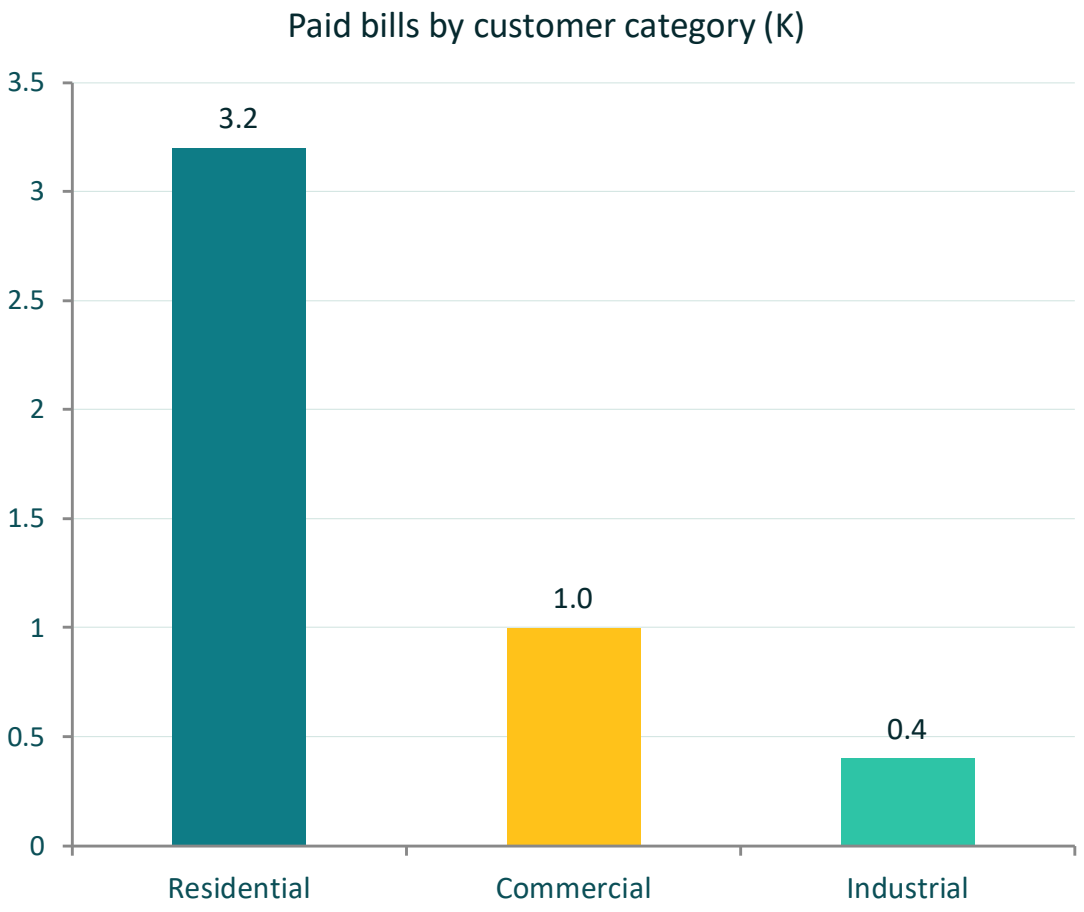
KEY KPIs

Unserved Power: **1.93K MW**
Total Demand: **5.42M MW**
Total Supply: **664.25 MW**
Outage Capacity Lost: **12.94K MW**
Avg Substation Load: **53.43%**

VISUALS

- Grid stress zone risk matrix
- Root cause waterfall and plant anomaly
- Outage impact: forced vs planned
- Digital twin map and substation health

Key Insights



- Availability reads 100% but PLF is 27.77%: capacity is ready but under-used.
- Grid frequency holds at 49.99 Hz, close to the 50 Hz nominal.
- Mundra UMTPP ranks first for lost capacity in the outage impact view.
- Residential customers account for the largest share of paid bills.
- Technical loss stands at 6.08% across the grid.
- Odisha has the highest value in the zone metric view, then Delhi and Mumbai.

Business Benefits

Faster decisions

One live view replaces scattered reports

Early risk detection

Outages, overloads and alerts surface quickly

Revenue clarity

Billing, collections and EV income in one place

Cleaner energy tracking

Generation mix and CO2 monitored together

EV growth insight

Charging sessions and peak hours by zone

Single source of truth

One MySQL model feeds every page

Conclusion & Future Scope

Conclusion

- MySQL and Power BI work together as one 360° control centre
- 16 tables feed 4 dashboards covering plants, grid, billing, EV and risk
- Clean data and DAX measures give trusted KPIs
- The model is ready for live DirectQuery refresh

Future Scope

- Connect to real-time plant and grid feeds
- Predict equipment faults with machine learning
- Forecast demand and solar output with AI
- Send automatic alerts by email or Teams
- Add a mobile view for field teams

Thank You

Questions & Discussion

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